

The constant $A(p)$

Quasi-stationary amplitude, closed forms, and the Koenigs obstruction

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Chapter 3

The constant $A(p)$

3.1 Introduction

For the binary Galton–Watson process with division probability $p \in [0, \frac{1}{2})$, the survival probability satisfies $S_n \sim A(p)(2p)^n$, so the limiting mean population size conditional on non-extinction is $1/A(p)$. Chapter 2 (*Mathematical introduction*) introduced this constant, derived the conditional mean and variance it controls, and recorded its continuous-time counterpart $A_c(p) = (1 - 2p)/(1 - p)$, which is elementary. The discrete constant is not elementary, and this chapter is devoted to everything that can nevertheless be said about it exactly.

The results of this chapter are original to this thesis. They comprise an exact infinite-product representation of $A(p)$ (section 3.3); an exact series with two-sided bounds that pin $A(p)$ between $\frac{1}{2}A_c(p)$ and $2\varepsilon/(1 + 2\varepsilon)$ (section 3.4); a rigorous near-critical asymptotic with remainder, $A(p) = 2\varepsilon(1 + O(\varepsilon \ln(1/\varepsilon)))$ for $\varepsilon = 1 - 2p$ (section 3.5); the resolution of the discrepancy between the discrete and continuous constants (section 3.6); an account of symbolic-regression searches for an elementary formula, and of why they failed (section 3.7); the identification of $A(p)$ with twice the Koenigs linearising function of the logistic map at $w = \frac{1}{2}$ and multiplier $r = 2p$ (section 3.8); and the dynamical obstruction that explains the failure of the searches (sections 3.9 and 3.10). Drawing on the classification of differentially algebraic Schröder functions due to Becker and Bergweiler [1], section 3.9 proves that for every $r \in (0, 1)$ the Koenigs function ψ_r is hypertranscendental in its own variable — with no exceptional parameters at all in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points. The classical elementary cases $r = 2$ and $r = 4$, both repelling and both outside the branching range, are derived in section 3.A and located inside the classification.

The scope of the obstruction is stated with care in section 3.10, because it is easy to overstate. The theorem constrains the conjugacy as a function of its dynamical variable z , and licenses no conclusion about individual values $A(p_0)$ — transcendence, and even irrationality, remain open — nor about the map $p \mapsto A(p)$, whose elementarity and differential algebraicity in p are likewise open. High-precision integer-relation searches at eleven rational parameters, described in section 3.10, supply the numerical evidence for the conjecture that no closed form exists under any reading; they are finite evidence, and are presented as such. Section 3.11 closes with the hybrid product/asymptotic scheme used to evaluate $A(p)$ and $1/A(p)$ throughout the rest of this thesis.

3.2 Standing setup

This section fixes the standing assumptions and notation for the chapter. The branching background is summarised only as far as the constant $A(p)$ requires; the full development of the process, of its critical behaviour, and of the quasi-stationary distribution whose mean $A(p)$ controls, is in Chapter 2 (*Mathematical introduction*), and the reader is referred there for proofs of everything quoted here without one.

3.2.1 The binary Galton–Watson process

Throughout, Z_n denotes the size of the n th generation of a Galton–Watson process started from a single progenitor, $Z_0 = 1$, in which each individual independently either divides into two, with probability p , or dies, with probability $1 - p$:

$$\mathbb{P}(L = 2) = p, \quad \mathbb{P}(L = 0) = 1 - p, \quad (3.1)$$

L being the offspring variable. The offspring probability generating function is $\phi(z) = pz^2 + (1 - p)$, the mean offspring number is $2p$, and

$$\mathbb{E}(Z_n) = (2p)^n. \quad (3.2)$$

Writing $u_n = \mathbb{P}(Z_n = 0)$ for the probability of extinction by generation n and $S_n = 1 - u_n = \mathbb{P}(Z_n > 0)$ for the probability of survival to generation n , the standard first-step argument [2, 3] gives $u_{n+1} = \phi(u_n)$, $u_0 = 0$, equivalently

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (3.3)$$

The sequence S_n is non-increasing and bounded below by 0, hence convergent. For $p \leq \frac{1}{2}$ the only fixed point of (3.3) in $[0, 1]$ is the origin, so extinction is certain; the case of interest in this chapter is the open subcritical interval.

Standing assumption. $0 \leq p < \frac{1}{2}$, with the endpoint convention $A(0) := \lim_{p \downarrow 0} A(p) = \frac{1}{2}$ adopted in section 3.4 once the limit is available.

3.2.2 The constant $A(p)$ and the quasi-stationary mean

Fix $0 < p < \frac{1}{2}$. Since $S_n \rightarrow 0$, the quadratic term in (3.3) becomes negligible beside the linear one, and the recursion is increasingly well approximated by $S_{n+1} \approx 2pS_n$. The decay is therefore geometric at rate $2p$, with a prefactor recording everything the linearisation forgets:

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad S_n \sim A(p)(2p)^n \quad (n \rightarrow \infty). \quad (3.4)$$

That the limit exists, and that $A(p)$ lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$, is proved in section 3.3 by way of the infinite product (3.8) below; for the moment $A(p)$ is the object whose properties this chapter develops.

Decomposing the expectation into surviving and extinct realisations,

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n} = \frac{(2p)^n}{S_n} \xrightarrow{n \rightarrow \infty} \frac{1}{A(p)}. \quad (3.5)$$

The geometric factors cancel exactly, so the surviving population has a constant limiting mean, the *quasi-stationary mean* $1/A(p)$; the limiting conditional variance, and the convergence of the full conditional distribution in the sense of Yaglom [4], are established in Chapter 2 (*Mathematical introduction*). The constant $A(p)$ is therefore the single quantity on which the late-time statistics of the subcritical process depend, and computing it — exactly where possible, rigorously otherwise — is the business of this chapter.

3.3 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2pw_n(1 - w_n) = r w_n(1 - w_n), \quad r = 2p, \quad w_0 = \frac{1}{2}, \quad (3.6)$$

which is the logistic map — the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 (figure 3.1). The subcritical window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is the substance of section 3.9.

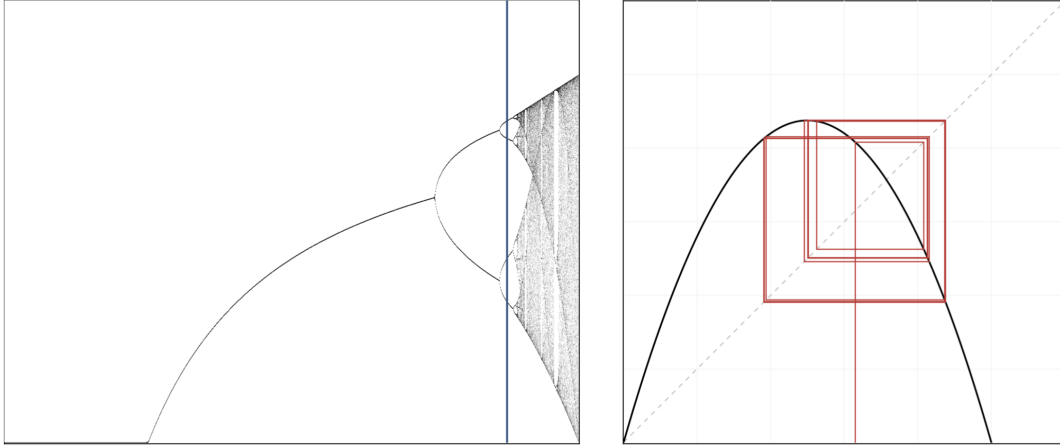


Figure 3.1: Bifurcation diagram of the logistic map $w \mapsto rw(1 - w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r . The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

In terms of w , (3.4) reads

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (3.7)$$

Any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0} \right) \left(\frac{x_2}{x_1} \right) \cdots \left(\frac{x_n}{x_{n-1}} \right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \quad \text{so} \quad \lim_{n \rightarrow \infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k},$$

and applying this with $x_n = w_n/r^n$ the ratio simplifies at once:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{rw_k(1-w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

Hence, with $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$,

$$A(p) = 2 \cdot \frac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k),$$

and pulling out the factor $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (3.8)$$

3.3.1 Convergence, and existence of the limit

For $0 < w_k < 1$ the infinite product $\prod_{k \geq 1} (1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (3.6) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$; since $0 < r < 1$ this bound is summable, and the product converges to a positive finite value. That also establishes what was asserted in [section 3.2.2](#): the limit (3.4) defining $A(p)$ exists and lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

The finite partial products

$$A_N(p) = \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right) \quad (3.9)$$

decrease monotonically to $A(p)$, so every truncation is a rigorous upper bound. A useful lower bound accompanies it: since $\prod_k (1 - a_k) \geq 1 - \sum_k a_k$ for $a_k \in [0, 1]$, and since $w_{k+1} \leq rw_k$ gives $\sum_{k > N} w_k \leq w_{N+1}/(1 - r)$,

$$A_N(p) \left(1 - \frac{w_{N+1}}{1 - r}\right) \leq A(p) \leq A_N(p) \quad (3.10)$$

whenever the left factor is positive. The bracket provides a certified stopping criterion for numerical evaluation: it is stronger than merely observing that successive iterates have stopped changing, and it is what [section 3.11](#) uses to decide when enough factors have been taken.

3.3.2 Is this an advance?

Only a modest one. (3.8) is exact and its partial products decrease monotonically, so truncating gives a rigorous upper bound on $A(p)$ at every order, which the raw ratio $S_n/(2p)^n$ does not. But computing the factors still requires iterating the nonlinear map to generate the S_k , so the near-critical difficulty is untouched: as $p \rightarrow \frac{1}{2}^-$ the number of factors needed grows without bound. What the product does supply is a route to the sharper result of the next section.

3.4 An exact series and two-sided bounds

The product converts into a telescoping sum, and the sum is much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = r S_n \left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \rightarrow 1/A(p)$ by (3.4). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n(1 - \frac{S_n}{2})} - \frac{r^n}{S_n} = \frac{r^n}{S_n} \left[\frac{1}{1 - \frac{S_n}{2}} - 1 \right] = \frac{r^n}{S_n} \cdot \frac{S_n/2}{1 - \frac{S_n}{2}} = \frac{r^n}{2 - S_n}.$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 3.1 (Series representation). *For $p \in [0, \frac{1}{2})$, with S_n generated by (3.3),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (3.11)$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$. That is what makes it useful, because S_n is confined to $(0, 1]$ and therefore $2 - S_n$ to $[1, 2)$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 3.2 (Two-sided bounds). *With $\varepsilon = 1 - 2p$, for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}, \quad (3.12)$$

that is,

$$\frac{1 - 2p}{2(1 - p)} < A(p) < \frac{2(1 - 2p)}{3 - 4p}. \quad (3.13)$$

The lower bound is exactly half the continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ derived in Chapter 2 (Mathematical introduction):

$$A(p) > \frac{1}{2}A_c(p). \quad (3.14)$$

Proof. Since $0 < S_n \leq 1$ for all n we have $1 \leq 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2 - S_n) \leq r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \geq 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon},$$

and inverting gives (3.12). Substituting $\varepsilon = 1 - 2p$ gives (3.13), whose lower bound is $(1 - 2p)/(2(1 - p)) = \frac{1}{2}A_c(p)$. $\square$

Two features of [proposition 3.2](#) deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \rightarrow 0$, we have $\varepsilon \rightarrow 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \leq \frac{1}{2}$ by [proposition 3.3](#) below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$. At the critical end, $p \rightarrow \frac{1}{2}^-$, we have $\varepsilon \rightarrow 0$, the lower bound behaves like ε and the upper like 2ε ; [section 3.5](#) shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, which is why no constant rescaling of A_c can reproduce it.

Proposition 3.3 (Parity bound). *For the binary Galton–Watson process started from a single individual, $A(p) \leq \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \rightarrow 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \geq 1$. Conditional on survival, therefore, $Z_n \geq 2$, and hence $\mathbb{E}(Z_n \mid Z_n > 0) \geq 2$ for all $n \geq 1$. Letting $n \rightarrow \infty$ and using (3.5) gives $1/A(p) \geq 2$. As $p \rightarrow 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit; combined with the lower bound of proposition 3.2, this proves the endpoint limit $A(p) \rightarrow \frac{1}{2}$ as $p \downarrow 0$. $\square$

The definition is therefore extended continuously by

$$A(0) := \lim_{p \downarrow 0} A(p) = \frac{1}{2}, \quad (3.15)$$

the convention advertised in section 3.2, which avoids the undefined ratio $S_n/(2p)^n$ at $p = 0$ itself. Figure 3.2 displays the squeeze of proposition 3.2 across the whole interval.

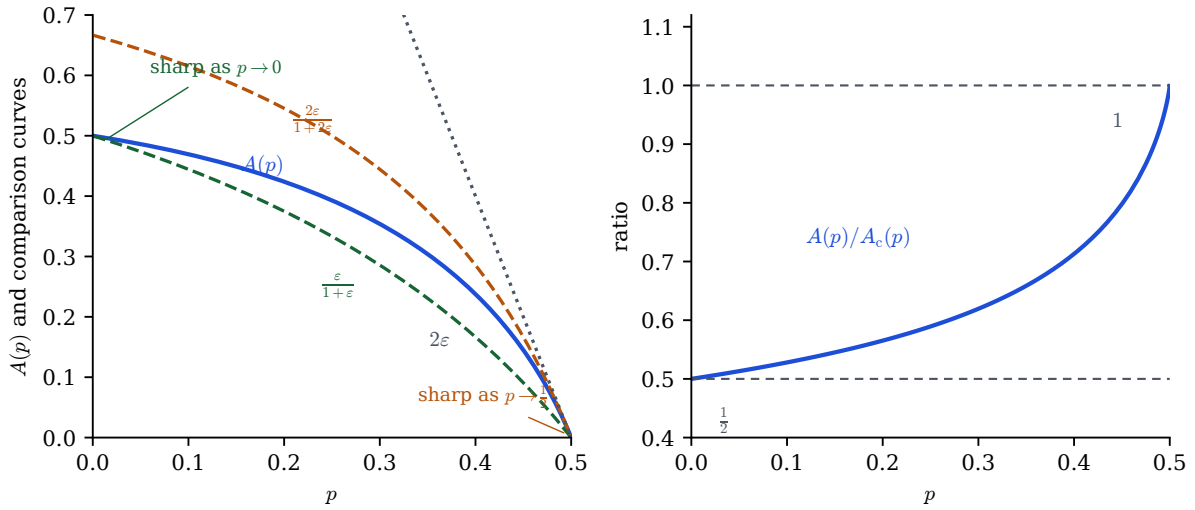


Figure 3.2: Left: the constant $A(p)$ (solid, from the exact series of proposition 3.1) between its two-sided bounds $\varepsilon/(1+\varepsilon)$ and $2\varepsilon/(1+2\varepsilon)$ of (3.12), with the near-critical asymptote 2ε dotted. The lower bound is attained at $p = 0$, the upper at $p = \frac{1}{2}$, and $A(p)$ runs from one to the other across the interval. Right: the ratio $A(p)/A_c(p)$, monotone from its endpoint limits $\frac{1}{2}$ and 1 of (3.20); no constant rescaling of the continuous-time constant reproduces the discrete one.

3.5 Behaviour as $p \rightarrow \frac{1}{2}^-$

The bounds of (3.12) show that $A(p)$ is of order ε , but leave a factor of two unresolved. The series (3.11) determines the leading constant and, with a little more care, the order of the first relative correction.

The idea is visible before any estimates are made. Fix ε small. The summand $r^n/(2 - S_n)$ involves two decays on very different scales. The geometric factor $r^n = (1 - \varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large. The survival probability S_n , by contrast, is close to its critical behaviour $S_n \sim 2/n$ over that whole range, and so has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is

Table 3.1: The constant $A(p)$ computed from the product (3.8) in 60-digit arithmetic, against the bounds of proposition 3.2 and the near-critical asymptotic $2(1 - 2p)$ of section 3.5. The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 3% of A , whereas at $p = 0.4999$ it is the upper bound that is within 0.1%. The last two columns give the quasi-stationary mean and the number of factors of (3.8) needed for the product to reach full precision — the cost that section 3.11 has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\frac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

approximately $r^n/2$. Summing,

$$\sum_{n=0}^{\infty} \frac{r^n}{2 - S_n} \approx \frac{1}{2} \sum_{n=0}^{\infty} r^n = \frac{1}{2\varepsilon},$$

so $1/A(p) \sim 1/(2\varepsilon)$ and

$$\boxed{A(p) \sim 2\varepsilon = 2(1 - 2p) \quad \text{as } p \rightarrow \frac{1}{2}^-} \quad (3.16)$$

Equivalently $1/A(p) \sim 1/(2(1 - 2p))$: the quasi-stationary mean diverges like the reciprocal of the distance from criticality. In particular $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4 , a fact used repeatedly below as a test of candidate formulae. The heuristic is now made rigorous, with the order of the remainder.

Theorem 3.4 (Near-critical asymptotic). *As $p \uparrow \frac{1}{2}$, with $\varepsilon = 1 - 2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \ln \frac{1}{\varepsilon}\right) \right]. \quad (3.17)$$

In particular (3.16) holds, and $1/A(p) \sim 1/(2(1 - 2p))$.

Proof. Write $r = 1 - \varepsilon$. The identity

$$\frac{1}{2 - S_n} = \frac{1}{2} + \frac{S_n}{2(2 - S_n)}$$

turns (3.11) into

$$\frac{1}{A(p)} = 1 + \frac{1}{2\varepsilon} + D_\varepsilon, \quad D_\varepsilon := \sum_{n=0}^{\infty} r^n \frac{S_n}{2(2 - S_n)}. \quad (3.18)$$

Only a logarithmic bound on the remainder is needed. The map $s \mapsto ps(2-s)$ is increasing in both p and s on the relevant range, so the subcritical survival probability is bounded above by its critical counterpart, $S_n(p) \leq S_n(\frac{1}{2})$. At $p = \frac{1}{2}$ the recursion gives

$$\frac{1}{S_{n+1}(\frac{1}{2})} - \frac{1}{S_n(\frac{1}{2})} = \frac{1}{2(1 - S_n(\frac{1}{2})/2)} \geq \frac{1}{2},$$

and summing the increments yields $S_n(\frac{1}{2}) \leq 2/(n+2)$. Since $2 - S_n \geq 1$,

$$\begin{aligned} 0 \leq D_\varepsilon &\leq \frac{1}{2} \sum_{n=0}^{\infty} r^n S_n(p) \leq \sum_{n=0}^{\infty} \frac{r^n}{n+2} \\ &= \frac{-\ln(1-r) - r}{r^2} = O\left(\ln \frac{1}{\varepsilon}\right). \end{aligned} \tag{3.19}$$

Factoring $1/(2\varepsilon)$ from (3.18) gives

$$\frac{1}{A(p)} = \frac{1}{2\varepsilon} \left[1 + 2\varepsilon \{1 + D_\varepsilon\} \right],$$

in which the quantity after the leading one is $O(\varepsilon \ln(1/\varepsilon))$; inverting proves (3.17). $\square$

The proof locates the logarithm that slows convergence to the leading asymptotic. The same result can also be reached through the continuum Riccati equation for the survival recursion, and it is worth seeing why that route is the worse one. The exact difference is

$$S_{n+1} - S_n = (2p-1)S_n - pS_n^2 = -\varepsilon S_n - \frac{1}{2}S_n^2 + \frac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $dS/dn = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = Ce^{\varepsilon n} - \frac{1}{2\varepsilon}, \quad S(n) = \frac{1}{Ce^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim Ae^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (3.16). Both steps are heuristic, and the second is delicate; the series argument is preferable because it needs no second matching at all.

Remark 3.5 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (3.8) in high precision gives $A/2\varepsilon = 0.90987, 0.98612, 0.99814, 0.99977$ and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6} . Fitting the deficit gives

$$1 - \frac{A}{2\varepsilon} \approx \varepsilon \left(\ln \frac{1}{\varepsilon} + c \right)$$

with $c \approx 0.78$, though the constant is empirical and is not used in what follows; the proved statement is the order estimate of [theorem 3.4](#). The logarithmic correction is why (3.16) is a good practical surrogate only very close to criticality, and it is what fixes the crossover point chosen in [section 3.11](#). [Figure 3.3](#) shows the correction over five decades of ε .

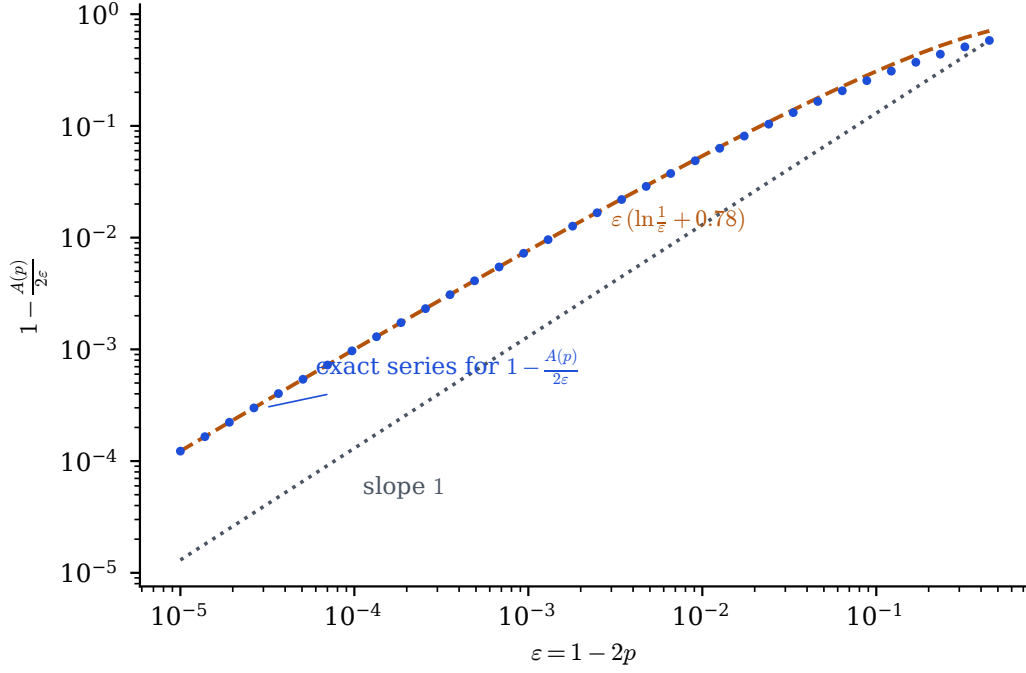


Figure 3.3: The relative deficit $1 - A(p)/2\varepsilon$ of the near-critical asymptotic, computed from the exact series of [proposition 3.1](#) (points), against $\varepsilon = 1 - 2p$ on logarithmic axes. The dotted line has slope 1; the departure of the points from it is the logarithmic correction of [remark 3.5](#), tracked by the dashed empirical fit $\varepsilon(\ln \frac{1}{\varepsilon} + 0.78)$. The deficit is of order $\varepsilon \ln(1/\varepsilon)$, exactly the remainder of [theorem 3.4](#), which is why the leading asymptotic alone is accurate only within a few parts in 10^3 of criticality.

3.6 Why the discrete and continuous constants differ

The two quasi-stationary constants now in hand are

$$A_c(p) = \frac{1 - 2p}{1 - p} \quad (\text{continuous time}), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \quad (\text{discrete time}),$$

and the observation that organises this section is that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves ([figure 3.4](#)).

[Proposition 3.2](#) explains this completely. What holds is not $A = \frac{1}{2}A_c$ but the strict inequality $A > \frac{1}{2}A_c$, and the inequality is attained only in the limit $p \rightarrow 0$. Combining [\(3.16\)](#) with the elementary form of A_c , the ratio runs across the interval with the endpoint limits

$$\frac{A(p)}{A_c(p)} \longrightarrow \frac{1}{2} \quad (p \rightarrow 0), \quad \frac{A(p)}{A_c(p)} \longrightarrow 1 \quad (p \rightarrow \tfrac{1}{2}^-), \quad (3.20)$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. The computed ratios increase monotonically through this list; that monotonicity is descriptive, and no monotonicity theorem is needed for [\(3.20\)](#). No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

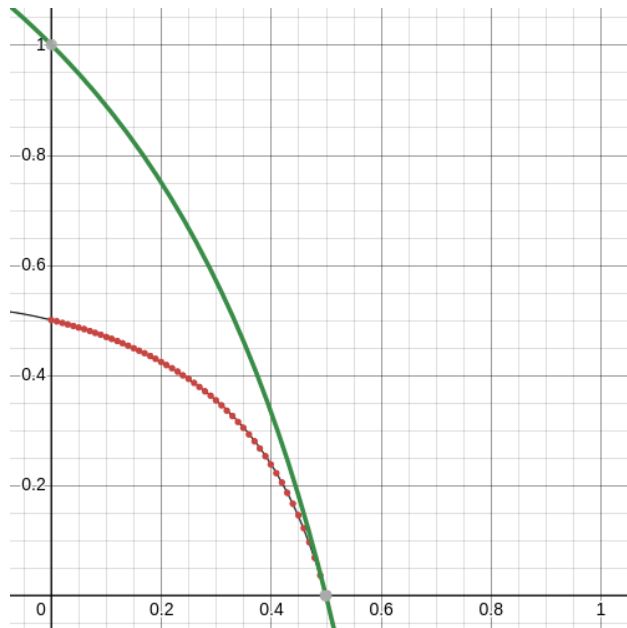


Figure 3.4: The elementary continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is one of the fitted surrogates produced by the symbolic-regression search of [section 3.7](#); it tracks the points closely but is not exact.

At $p \rightarrow 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even ([proposition 3.3](#)); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \rightarrow \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding $A_c(p) = (1 - 2p)/(1 - p)$ near $p = \frac{1}{2}$ gives $A_c(p) \sim 2(1 - 2p)$, which is precisely [\(3.16\)](#). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order.

The discrete curve is therefore squeezed between $\frac{1}{2}A_c$ and its own critical asymptote, hugging the former at one end and the latter at the other. That is the whole content of [figure 3.4](#).

One point of comparison deserves emphasis, because it is what makes the discrete constant hard and the continuous one easy. The two parametrisations are matched by the competing-clocks identity $p = \lambda/(\lambda + \mu)$, proved in Chapter 2 (*Mathematical introduction*) and not repeated here: the continuous-time constants arise from a linear differential equation whose coefficients factorise, while $A(p)$ retains the cumulative effect of every nonlinear iteration of [\(3.3\)](#). The continuous constant can be read off the late-time equation alone; the discrete one cannot, and the product, series and asymptotic of the preceding sections are what stand in its place.

3.7 Searching for a closed form

Before the connection of [section 3.8](#) was appreciated, several methods were tried in the hope of an elementary formula for $A(p)$. The negative result is worth recording, and so is the manner of the failure.

The first method was the continuum route discussed in [section 3.5](#): treat the map as a differential equation in the large-generation limit and solve it. This gives the shape of the decay but leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The idea is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [\[5, 6\]](#). The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \dots, a_k]$ for $a_0 + a_1p + \dots + a_kp^k$ up to some maximum order, drawn uniformly at random. It became apparent quickly that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched $\sum_i (a_i/b_i)p^i$ with integer $[a_i], [b_i]$, and a third extended the space to ratios of two such polynomials. The best that emerged merely fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (3.21)$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing, but it fails the diagnostic of [\(3.16\)](#): differentiating at $p = \frac{1}{2}$ gives a gradient of $-2/0.55 = -3.636$ rather than the required -4 .

A later attempt used a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when candidates with fewer components are explicitly favoured. The expressions returned were typically deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, and they were not interpretable in any useful sense; representative candidates $\hat{A}_2$, $\hat{A}_3$ and $\hat{A}_4$, with their plots, are quarantined in [section 3.B](#) as records of the fitting exercise rather than proposed identities.

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4 , and near criticality this matters enormously: as $A \rightarrow 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve, forcing the correct critical branch; the construction is recorded in [section 3.B](#), and it does work. But if one is going to splice, it is better to splice onto something exact, and [section 3.11](#) does so.

The deeper lesson points directly at the next section. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve fitting cannot distinguish a function that has a closed form from one that does not.

3.8 Identification with the Koenigs function

Some time after the search of [section 3.7](#) had been abandoned, a dynamical argument emerged that explains why no elementary closed form should be expected. In short: $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range required. This section makes the identification precise; the next supplies the obstruction.

3.8.1 Koenigs linearisation

Definition 3.6 (Koenigs function). Let $f_r(z) = rz(1-z)$ have a fixed point at $z = 0$ with multiplier $r = f'_r(0)$ satisfying $0 < |r| < 1$. Koenigs [7] proved that there is a unique function ψ_r , holomorphic in a neighbourhood of the origin, satisfying the Schröder equation

$$\psi_r(f_r(z)) = r \psi_r(z), \quad \psi_r(0) = 0, \quad \psi'_r(0) = 1. \quad (3.22)$$

Equivalently,

$$\psi_r(z) = \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(z) \quad (3.23)$$

uniformly near the origin. The normalisation $\psi'_r(0) = 1$ is equivalent to $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$.

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ_r coordinate the nonlinear map f_r becomes multiplication by r ([figure 3.5](#)). Standard accounts are Milnor [8] and Carleson and Gamelin [9]. Throughout this chapter $f_r(z) = rz(1-z)$ is exactly (3.6), with $z = w_n$ and $f_r(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting.

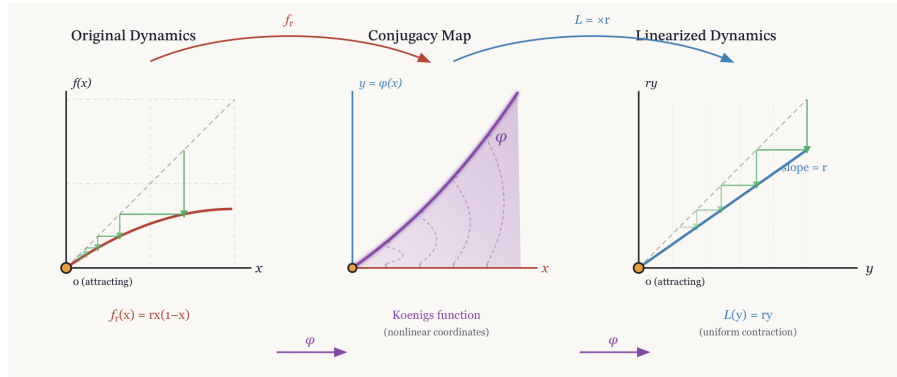


Figure 3.5: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

3.8.2 The identity $A(p) = 2\psi_r(\frac{1}{2})$

The local Koenigs function is initially defined only in a neighbourhood of 0. For $r \in (0, 1)$ the origin is attracting on the real line and the interval $(0, 1)$ lies in its real basin: starting from $w_0 = \frac{1}{2}$, one has $w_n \rightarrow 0$. On that basin the limit

$$\Psi_r(w_0) := \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(w_0) \quad (3.24)$$

exists and agrees with the local conjugacy near 0, since the normalisation $\psi'_r(0) = 1$ gives $\psi_r(w_n)/r^n \sim w_n/r^n$ along the orbit.

Proposition 3.7. *For $p \in (0, \frac{1}{2})$ and $r = 2p$,*

$$A(p) = 2\Psi_r\left(\frac{1}{2}\right) = 2 \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}\left(\frac{1}{2}\right). \quad (3.25)$$

Proof. Along the orbit, whenever w_n lies in the local domain, iterating (3.22) gives $\psi_r(w_1) = r\psi_r(w_0)$, hence $\psi_r(w_0) = \psi_r(w_1)/r$, and by induction

$$\psi_r(w_0) = \frac{\psi_r(w_n)}{r^n} \quad \text{for every } n. \quad (3.26)$$

Passing to the limit, $\psi_r(w_0) = \lim_n \psi_r(w_n)/r^n$. Since $w_n \rightarrow 0$ and $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$, replacing $\psi_r(w_n)$ by w_n in the limit is legitimate, and $\psi_r(w_0) = \lim_n w_n/r^n$. Comparing with (3.7) and using $w_n = S_n/2$ and $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\Psi_r(\frac{1}{2})$. $\square$

We write $A(p) = 2\psi_r(\frac{1}{2})$ as a shorthand for this basin value of the Koenigs conjugacy, not as a naive evaluation of a power series beyond its guaranteed disc of convergence; the relation between the germ and the basin is taken up in [remark 3.16](#). Thus every question about an elementary closed form for $A(p)$ is tied to the nature of that conjugacy at multiplier $r = 2p$.

3.8.3 Conjugacy with the quadratic family

The logistic family is affinely conjugate to the canonical quadratic family, and the correspondence is used repeatedly below.

Lemma 3.8. *The logistic map $f_r(x) = rx(1 - x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with*

$$c = c(r) := \frac{2r - r^2}{4}. \quad (3.27)$$

Under this correspondence the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$.

Proof. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $x = -z/r + \frac{1}{2}$ (for $r \neq 0$) yields a monic quadratic; matching coefficients gives $c = (2r - r^2)/4$. Since $c(r)$ increases on $(0, 1)$, with $c(0) = 0$ and $c(1) = \frac{1}{4}$, the image of the branching window is $(0, \frac{1}{4})$. $\square$

The interval $c \in (0, \frac{1}{4})$ is the real diameter of the main cardioid of the Mandelbrot set: a fixed point of $z^2 + c$ with multiplier r has coordinate $z_* = r/2$, and its fixed-point equation gives $c = z_* - z_*^2 = r/2 - r^2/4$, with the endpoint $r \uparrow 1$ approaching the parabolic cusp $c = \frac{1}{4}$, which is criticality. This locates the parameter range geometrically; its role in the obstruction argument is fixed in the next section, and it is not used as evidence for any closed-form claim.

3.9 Hypertranscendence of the conjugacy

This section makes the obstruction to an elementary closed form precise, and delimits exactly what it does and does not assert. The logical skeleton is short: (i) every elementary function satisfies an algebraic differential equation; (ii) differential algebraicity survives compositional inversion; (iii) by a theorem of Becker and Bergweiler, a solution of Schröder's equation for a rational map of degree

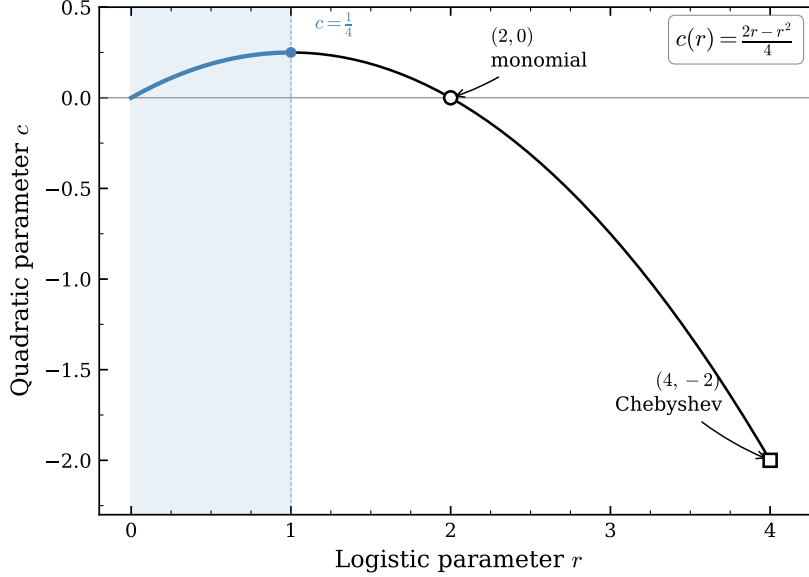


Figure 3.6: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1 - x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$. The two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it; see [section 3.A](#).

at least two can be differentially algebraic only when the underlying fixed point is *repelling*; (iv) for $r \in (0, 1)$ the fixed point 0 of f_r is attracting. Hypertranscendence of ψ_r for every subcritical parameter follows at once — with no exceptional cases to exclude, because the attracting window contains no candidates at all. We then explain, with equal care, why this theorem does *not* by itself settle the status of the individual numbers $A(p_0)$ or of the map $p \mapsto A(p)$, and we record the numerical evidence bearing on those.

3.9.1 Differential algebraicity, hypertranscendence, elementarity

Definition 3.9 (Differentially algebraic; hypertranscendental). A function g , holomorphic on a domain (or a formal power series), is *differentially algebraic over $\mathbb{C}(z)$* if there exists a non-zero polynomial P in $n + 2$ variables, for some $n \geq 0$, with

$$P(z, g(z), g'(z), \dots, g^{(n)}(z)) \equiv 0.$$

Otherwise g is *differentially transcendental*, or *hypertranscendental*.

Two closure facts make this the right notion for the present purpose.

Lemma 3.10 (Elementary $\Rightarrow$ differentially algebraic). *Every elementary function in the Liouville sense — any function reachable from $\mathbb{C}(z)$ by a finite tower of algebraic operations, exponentials and logarithms — is differentially algebraic. Consequently a hypertranscendental function is not elementary.*

Proof sketch. Each storey of a Liouvillian tower preserves differential algebraicity: algebraic extensions do so by elimination; if g is differentially algebraic then so are e^g and $\ln g$, their defining first-order relations combining with the equation for g ; and the class is closed under field operations and composition. These closure properties are classical; see Rubel’s survey [10]. $\square$

Lemma 3.11 (Closure under compositional inversion). *Let g be holomorphic near 0 with $g(0) = 0$ and $g'(0) \neq 0$, and let g^{-1} denote its local compositional inverse. Then g is differentially algebraic over $\mathbb{C}(z)$ if and only if g^{-1} is [11, 12].*

Lemma 3.11 matters because the literature states Schröder-equation results in two equivalent but opposite conventions, and conflating them is an easy way to misquote a theorem. Our ψ_r is the *linearising coordinate*: $\psi_r(f_r(z)) = r \psi_r(z)$. The function classified by Becker and Bergweiler (and by Ritt before them) is the *linearising parametrisation* $\varphi_r = \psi_r^{-1}$, which satisfies

$$\varphi_r(rt) = f_r(\varphi_r(t)), \quad \varphi_r(0) = 0, \quad \varphi_r'(0) = 1. \quad (3.28)$$

By lemma 3.11, hypertranscendence of either function transfers to the other.

3.9.2 The Becker–Bergweiler theorem, stated precisely

The result needed here is the 1995 classification of differentially algebraic Schröder functions [1], quoted in the form given in Fernandes’ survey of the Schröder, Böttcher and Abel equations [13, Thm 3.1].

Theorem 3.12 (Becker–Bergweiler [1]; as restated in [13, Thm 3.1]). *Let R be a rational map of degree at least two with $R(0) = 0$ and multiplier $s = R'(0)$, where $s \neq 0$ and s is not a root of unity (for $|s| = 1$ the Schröder function is understood as a formal series). Let φ , normalised by $\varphi(0) = 0$ and $\varphi'(0) = 1$, solve $\varphi(st) = R(\varphi(t))$. If φ is differentially algebraic over $\mathbb{C}(t)$, then:*

- (i) *the fixed point 0 is repelling, $|s| > 1$; and*
- (ii) *φ is a Möbius transformation of one of*

$$\exp(\alpha t^\rho), \quad \cos(\alpha t^\rho + \beta), \quad \wp(\alpha t^\rho + \beta), \quad \wp^2, \quad \wp^3, \quad \wp'(\alpha t^\rho + \beta),$$

where $\wp$ is a Weierstrass function, $\rho \in \mathbb{Q}$ is such that the function concerned is meromorphic on $\mathbb{C}$, $\alpha \neq 0$, and β is a fraction of a period. Correspondingly R is Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$ (Chebyshev), or to a Lattès map.

Remark 3.13 (Lineage and independent corroboration). For $|s| > 1$ — the Poincaré-function case — the classification goes back to Ritt in 1926 [14]; Becker and Bergweiler completed the picture for the three classical functional equations at once, and Bergweiler’s later account with Aschenbrenner [15] assigns the exceptional list explicitly to the repelling case. Di Vizio and Fernandes [16] have given an independent difference-Galois proof of Ritt’s theorem; notably, their Theorems 1.2 and 4.2 assume only that s is neither zero nor a root of unity — no condition $|s| > 1$ — and show that any differentially algebraic solution of $y(R(x)) = sy(x)$ (our ψ -convention, with no inversion needed) must already satisfy a first-order equation of Ritt type $(y')^\rho = B(x)y^j$. The structure underlying theorem 3.12 therefore does not hang on a single paper.

3.9.3 Hypertranscendence throughout the subcritical window

Theorem 3.14 (No exceptional parameters in $(0, 1)$). *For every $r \in (0, 1)$, the Koenigs function ψ_r of $f_r(z) = rz(1 - z)$ at 0 is hypertranscendental over $\mathbb{C}(z)$: it satisfies no algebraic differential equation. In particular, ψ_r is not an elementary function.*

Proof. Fix $r \in (0, 1)$. The map f_r is rational of degree two, fixes 0, and its multiplier $s = f'_r(0) = r$ satisfies $0 < s < 1$; the hypotheses of [theorem 3.12](#) hold, with 0 attracting. Suppose, for contradiction, that ψ_r were differentially algebraic. Its local inverse $\varphi_r = \psi_r^{-1}$ solves the parametrisation form (3.28) with the stated normalisation, and by [lemma 3.11](#) it would be differentially algebraic too. [Theorem 3.12\(i\)](#) would then force $|r| > 1$, contradicting $r \in (0, 1)$. Hence ψ_r is hypertranscendental, and non-elementarity follows from [lemma 3.10](#). $\square$

Remark 3.15 (The exceptional set inside the window is empty, not merely avoided). An earlier form of this argument reasoned through the parameter plane: the affine conjugacy $c(r) = (2r - r^2)/4$ of [lemma 3.8](#) sends $r \in (0, 1)$ to $c \in (0, \frac{1}{4})$, which avoids the exceptional quadratic parameters $c = 0$ (reached at $r = 2$) and $c = -2$ (reached at $r = 4$, and also at $r = -2$). That observation is true but logically superfluous, and it invites a question it cannot answer — *is the exceptional list, restricted to the window, complete?* The route through [theorem 3.12\(i\)](#) dissolves the question: the exceptional Schröder functions exist only over *repelling* fixed points, so the attracting window contains no candidates whatsoever, and no enumeration of exceptional parameters is required. Consistently, the classical elementary cases of [section 3.A](#) have multipliers 2 and 4 at the linearised fixed point — both repelling: they are Poincaré-type linearisations, and neither can be pressed into service for $r \in (0, 1)$.

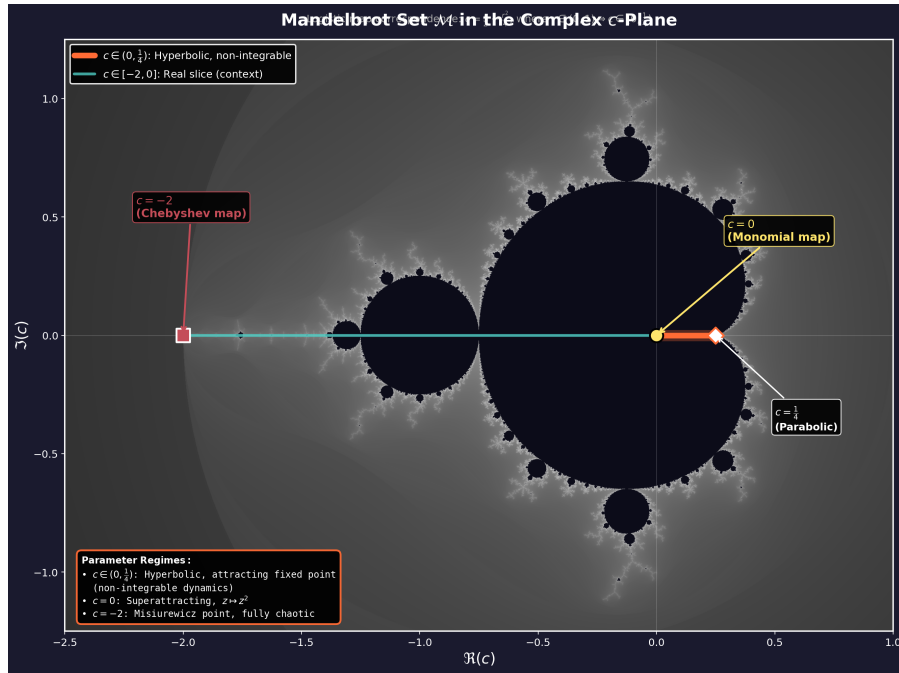


Figure 3.7: The Mandelbrot set in the complex c -plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of (3.27) this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$, which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. In the light of [remark 3.15](#) this picture is a consistency check on the theorem rather than part of the proof: the exceptional linearisations live over repelling fixed points, which the attracting window excludes outright. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$.

Remark 3.16 (Germ versus basin). [Theorem 3.14](#) concerns ψ_r as an analytic function; the quantity $A(p) = 2\Psi_r(\frac{1}{2})$ of [\(3.25\)](#) uses its extension to the real basin of attraction, since for $p \geq \frac{1}{3}$ the point $\frac{1}{2}$ lies outside the disc on which Koenigs' series is guaranteed to converge (radius $(1-r)/r$). The extension is by finitely many applications of the functional equation,

$$\Psi_r(w) = r^{-N} \psi_r(f_r^{\circ N}(w)) \quad \text{for } N \text{ large enough that } f_r^{\circ N}(w) \text{ enters the germ,}$$

and an algebraic differential equation, if one held on any subdomain, would persist under this analytic continuation. Hypertranscendence of the germ and of the basin extension are therefore equivalent statements, and the identity [\(3.25\)](#) is unaffected.

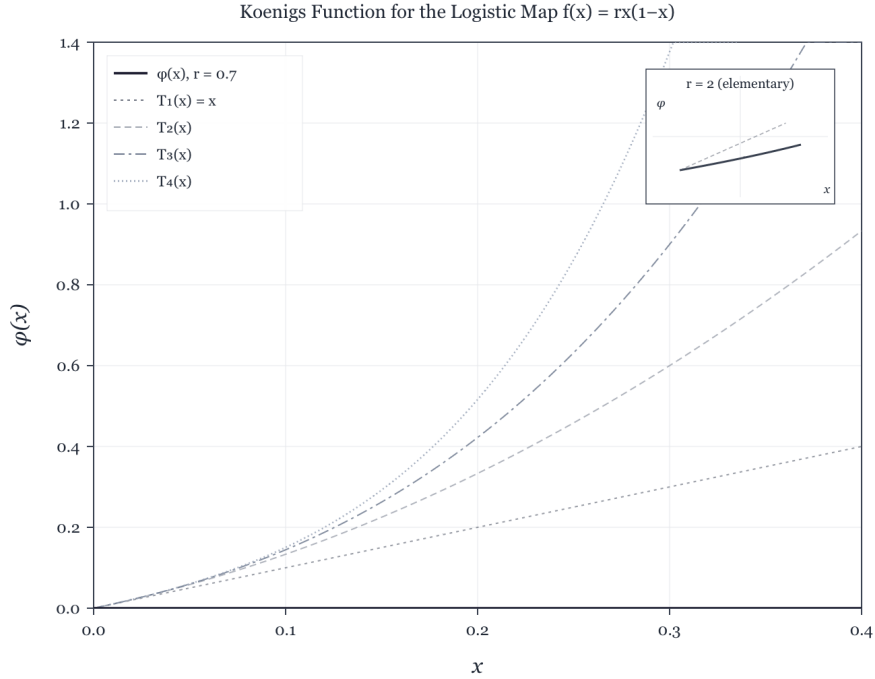


Figure 3.8: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range ([section 3.A](#)).

3.10 What the theorem does and does not say about $A(p)$

“Closed form” must itself be made precise before any claim is stated; at least four inequivalent readings are in play, and a result for one is not a result for another.

- (V) the *value* $A(p_0)$ is elementary or algebraic, for a fixed rational p_0 ;
- (E) the *map* $p \mapsto A(p)$ is an elementary function of p ;
- (D) the map is *differentially algebraic* in p , i.e. satisfies some algebraic ODE in p ;
- (S) the map is expressible via standard special functions (Γ , ζ , ${}_2F_1$, theta or q -series).

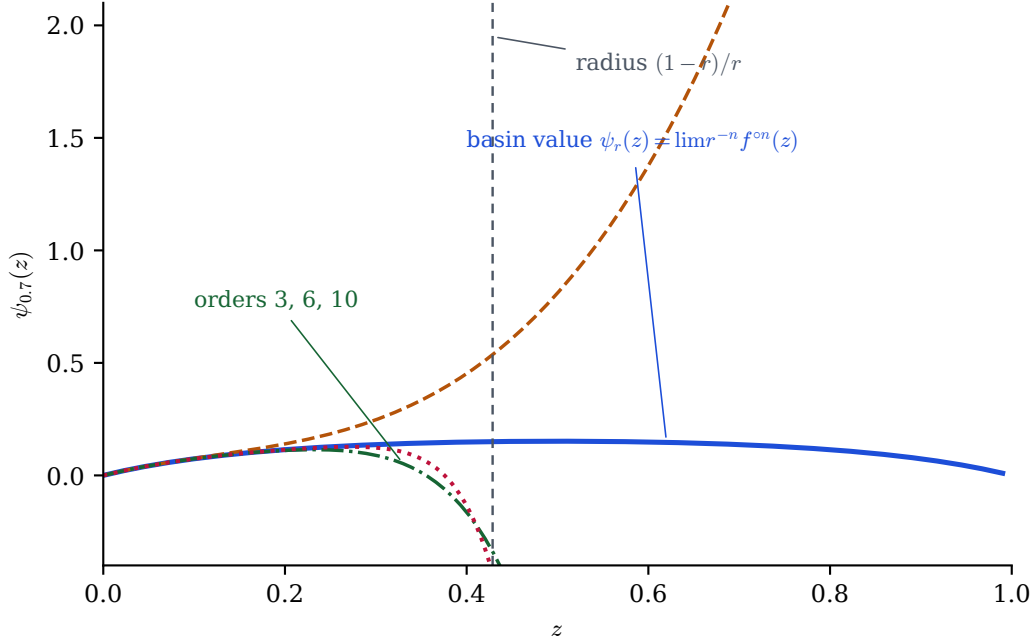


Figure 3.9: Germ versus basin for the Koenigs function at $r = 0.7$. The solid curve is the basin value $\psi_r(z) = \lim_n r^{-n} f_r^{on}(z)$, computed by direct orbit iteration and smooth on the whole of $[0, 1]$; the dashed curves are Taylor partial sums of orders 3, 6 and 10 about the origin, obtained by iterating the functional equation $\psi_r \circ f_r = r\psi_r$ on truncated series. The truncations agree with the basin value inside the guaranteed disc of radius $(1 - r)/r$ (vertical line) and diverge beyond it, the higher orders first — exactly the germ/basin distinction of [remark 3.16](#). Evaluating $A(p) = 2\Psi_r(\frac{1}{2})$ therefore requires the basin extension whenever $\frac{1}{2}$ lies outside the disc, that is for $p \geq \frac{1}{3}$.

[Theorem 3.14](#) speaks to the function $z \mapsto \psi_r(z)$ at fixed r , and licenses exactly this much: *no evaluation of $A(p)$ can be short-circuited through an algebraic differential equation for the lineariser, at any subcritical parameter.* It does not, by itself, decide any of (V), (E), (D) or (S). Two gaps must be stated plainly.

The value gap. A hypertranscendental function can take elementary — even algebraic — values at individual points: Γ is hypertranscendental by Hölder’s theorem, yet $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Nothing so far forbids, say, $A(\frac{1}{4})$ from being an algebraic number. There *is* a genuine values theorem in this circle of ideas: Becker and Bergweiler proved in 1993 [\[17\]](#) that the Böttcher conjugacy between two polynomials of the *same degree* $d \geq 2$ with algebraic coefficients (not linearly conjugate to monomials or Chebyshev polynomials) is transcendental and takes transcendental values at algebraic points; Mahler’s method drives the proof (for the method, see Nishioka [\[18\]](#)). One might hope to transfer this to the Koenigs value $\psi_r(\frac{1}{2})$. It does not transfer, for two independent reasons. *Formally*, the Schröder equation pairs an inner map of degree two with an outer map — multiplication by r — of degree one, so the equal-degree hypothesis fails outright. *Structurally*, no easy repair exists: Mahler’s method balances doubly-exponential analytic decay along an orbit against comparable height growth, and in the Böttcher setting both exponents run like d^n ; in the attracting Koenigs setting the orbit $w_n = f_r^{on}(\frac{1}{2})$ decays only geometrically, $|w_n| \sim A r^n$, while the heights of the rational numbers w_n still grow doubly exponentially, $h(w_n) \sim C 2^n$, and the resulting inequalities

run the wrong way at every n . Note that the obstruction is *not* the germ/basin issue — the pullback of [remark 3.16](#) moves the evaluation point arbitrarily deep into the germ at the cost of a rational factor — it is the degree structure of the equation itself. The consequence deserves emphasis:

transcendence — indeed, even irrationality — of $A(p_0)$ is at present not proved for a single rational $p_0 \in (0, \frac{1}{2})$.

The parameter gap. [Theorem 3.14](#) says nothing about the map $p \mapsto A(p)$: readings (E) and (D) are open. To the best of our knowledge the question is not merely unsolved but unaddressed in the literature. The natural machinery — the parametrised differential Galois theory of difference equations of Hardouin and Singer [\[19\]](#), designed precisely to detect differential relations in auxiliary directions for linear difference equations such as $\sigma\psi = r\psi$ — does not apply as it stands, because the derivation ∂_r fails to commute with the substitution operator $\sigma: w \mapsto f_r(w)$, which itself moves with r . Nor does any feature of the near-critical asymptotics obstruct (E) or (D): logarithmic corrections to [\(3.17\)](#) are themselves elementary functions of ε .

3.10.1 Inverse-symbolic evidence at eleven rational parameters

Since the theorems leave (V) open, the value question was attacked directly by integer-relation detection. The values

$$A(p), \quad p \in \left\{ \frac{1}{10}, \frac{1}{8}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{3}{10}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{5}{12}, \frac{9}{20} \right\},$$

were computed to more than 1150 significant digits by two mechanically independent routes — the exact product identity [\(3.8\)](#) iterated to convergence, and a pullback of the orbit into the Koenigs germ followed by evaluation of the Koenigs series — which agreed to at least 1192 digits at every parameter. Each value was then subjected to a PSLQ battery at ≥ 200 -digit effective rigour:

- *algebraicity*: no integer polynomial relation of degree ≤ 10 and coefficient height $\leq 10^6$;
- *linearity*: no rational linear relation, at height $\leq 10^8$, against the constants π , π^2 , $\ln 2$, $\ln 3$, $\ln 5$, γ , $\zeta(3)$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e , e^π , $\Gamma(\frac{1}{3})$, $\Gamma(\frac{1}{4})$ and Catalan’s constant (also tested for $1/A$, the quasi-stationary mean itself, and for A/ε);
- *bilinearity*: no representation $A = L_1/L_2$ with L_1, L_2 integer linear forms of height $\leq 10^8$ in a twelve-constant basis (run at 460 digits);
- *multiplicativity*: no relation $A = e^{q_0} 2^{q_1} 3^{q_2} 5^{q_3} 7^{q_4} \pi^{q_5}$ with rational exponents of height $\leq 10^{10}$;
- *cross-parameter*: no rational linear or multiplicative relation linking the eleven values to one another (run at 500 digits).

In all, 123 tests returned no relation — indeed no discovery-stage candidate: every PSLQ run terminated on its internal norm bound, so each null carries a genuine “no relation up to the stated height” guarantee. Throughout, the precision head-room (digits divided by basis size) exceeded every height cap by at least eight orders of magnitude, so a returned relation would have been meaningful rather than numerical noise.¹

¹Computations in `mpmath` at working precisions of 310–1200 digits; scripts, the 1150-digit values, and the full per-test log are archived with the project sources.

This is finite evidence, and its scope must be stated honestly: it excludes low-complexity closed forms over the tested constants, and nothing more. A closed form over untested constants ($\Gamma(\frac{1}{7})$, say, or periods of higher-genus curves), or one of height above the caps, would have escaped the search. It is evidence for the conjecture, not a proof.

3.10.2 The working claim

For every fixed $r = 2p \in (0, 1)$, the Koenigs linearisation ψ_r is hypertranscendental in its own variable (theorem 3.14) — a theorem, with no exceptional parameters in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points. Consequently no elementary expression in z for the conjugacy can be evaluated to produce $A(p)$. Whether individual values $A(p_0)$ are transcendental (or even irrational), and whether the map $p \mapsto A(p)$ is elementary or differentially algebraic in p , remain open questions; no closed form under any reading is known, none is expected, and eleven rational values resist inverse-symbolic identification against standard constants at over 200-digit rigour.

3.11 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes of section 3.4 suggest how to split the work.

Over most of the interval, direct evaluation of the infinite product (3.8) is entirely satisfactory: its partial products bound $A(p)$ from above at every order, and the tail bracket (3.10) bounds it from below, so every truncation carries a certified error bar rather than a hope of convergence. The bracket also supplies the stopping criterion: iterate until the bracket is narrower than the required tolerance, which is stronger than observing that successive floating-point iterates have stopped changing. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \rightarrow 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes an unbounded error in the quasi-stationary mean. Second, and this is the binding constraint, as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 3.1 shows what that costs: about 10^2 factors at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (3.17), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\ 2 - 4p, & p \geq p_*, \end{cases} \quad (3.29)$$

with the crossover p_* chosen just below $\frac{1}{2}$. Remark 3.5 indicates where to put it: the relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$ by theorem 3.4, so at $p_* = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3 , while the product still converges in a few hundred thousand iterations. Choosing $p_* = 0.499$ instead leaves an error of about 1.4% at the join. Any p_* in the range 0.499 to 0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

This is the recipe used for the numerical values quoted throughout this chapter, including those in [table 3.1](#); all of them are generated from the product rather than from a fitted formula, with the asymptotic switched on only inside the crossover window. For calculations that do not require a certified product value arbitrarily close to criticality, [\(3.29\)](#) restores accurate quasi-stationary means $1/A(p)$ even where raw iteration becomes expensive.

3.12 Conclusion

The constant $A(p)$ organises the quasi-stationary mean of subcritical binary Galton–Watson branching. It is well defined, computable, and linked to the basin Koenigs value $A(p) = 2 \lim_n r^{-n} f_r^{\circ n}(\frac{1}{2})$ at multiplier $r = 2p$. For every $r \in (0, 1)$ that conjugacy is hypertranscendental in its own variable ([theorem 3.14](#)) — a theorem with no exceptional parameters in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points — and this explains the failure of every elementary closed-form search of [section 3.7](#), while leaving $A(p)$ fully usable for analysis and numerics through the scheme of [section 3.11](#).

What the theorem does not decide has been flagged with equal care. Transcendence — even irrationality — of individual values $A(p_0)$ remains open, because the 1993 Becker–Bergweiler values theorem [\[17\]](#) is confined to equal-degree Böttcher pairs and does not transfer to the Schröder setting. The elementarity and differential algebraicity of the map $p \mapsto A(p)$, as opposed to the Koenigs function ψ_r in its argument, are likewise open; the former question appears nowhere in the literature, and the natural machinery of parametrised differential Galois theory [\[19\]](#) does not apply as it stands. Eleven rational values nonetheless resist all inverse-symbolic identification at over 200-digit rigour, which is what the evidence of [section 3.10.1](#) amounts to — and no more.

The exceptional elementary linearisations at $r = 2$ and $r = 4$, both repelling, lie outside the branching window; their derivations, and their place inside the Becker–Bergweiler classification, are collected in [section 3.A](#), and the exploratory formulae of the closed-form search in [section 3.B](#). The open problems proper to this chapter are therefore three, corresponding to the readings of [section 3.10](#): the arithmetic nature of the values $A(p_0)$ at rational p_0 ; the elementarity and differential algebraicity of the parameter map $p \mapsto A(p)$; and its expressibility, if any, in terms of standard special functions.

3.A Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching ($r = 2p < 1$) but complete the classical picture of integrable logistic linearisations. Both are recorded in full because they show concretely what a solvable case looks like, and because they sit exactly where the Becker–Bergweiler classification, [theorem 3.12](#), says all differentially algebraic Schröder functions must sit: over repelling fixed points.

3.A.1 Case $r = 2$ (logarithmic)

For $r = 2$, $f_2(x) = 2x(1 - x)$. The fixed point 0 has multiplier $\lambda = 2$ (repelling); the linearisation is of Poincaré type rather than an attracting-Koenigs function, but an elementary closed form still exists.

The affine change $u = 1 - 2x$ conjugates the dynamics to $u \mapsto u^2$, since

$$u_{n+1} = 1 - 2(2x_n(1 - x_n)) = 1 - 4x_n + 4x_n^2 = (1 - 2x_n)^2 = u_n^2.$$

Linearising $u \mapsto u^2$ by the logarithm, since $\ln(u^2) = 2 \ln u$, and returning to x gives the candidate

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \quad (3.30)$$

normalised so that $\psi_2(0) = 0$ and $\psi_2'(0) = 1$. Indeed,

$$\psi_2(f_2(x)) = -\frac{1}{2} \ln(1 - 4x + 4x^2) = -\frac{1}{2} \ln((1 - 2x)^2) = 2\psi_2(x).$$

3.A.2 Case $r = 4$ (inverse trigonometric)

For $r = 4$, $f_4(x) = 4x(1 - x)$. The fixed point 0 has multiplier $f_4'(0) = 4$ (repelling); as with $r = 2$, this is a Poincaré-type linearisation, not an attracting Koenigs function. With $x = \sin^2 \theta$ one finds

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The candidate

$$\psi_4(x) = (\arcsin \sqrt{x})^2 \quad (3.31)$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ on a suitable real domain: using $\arcsin(2u\sqrt{1-u^2}) = 2 \arcsin u$ with $u = \sqrt{x}$,

$$\psi_4(f_4(x)) = (\arcsin \sqrt{4x(1-x)})^2 = (2 \arcsin \sqrt{x})^2 = 4\psi_4(x).$$

And as $x \rightarrow 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi_4'(0) = 1$.

3.A.3 Location inside the classification

Under the affine conjugacy of lemma 3.8, $r = 2 \Leftrightarrow c = 0$ (monomial z^2) and $r = 4 \Leftrightarrow c = -2$ (Chebyshev type; the value $c = -2$ is also reached at $r = -2$). Neither exceptional value lies in the image of $r \in (0, 1)$, which is $c \in (0, \frac{1}{4})$. As explained in remark 3.15, this parameter count is a consistency check rather than the proof: the decisive point is that both exceptional cases sit over *repelling* fixed points, which the attracting window excludes structurally.

The two derivations above realise the first two families of theorem 3.12 exactly. For $r = 2$, inverting (3.30): $t = -\frac{1}{2} \ln(1 - 2x)$ gives

$$\varphi_2(t) = \psi_2^{-1}(t) = \frac{1 - e^{-2t}}{2},$$

an affine (hence Möbius) image of e^{-2t} — the family $\exp(\alpha t^\rho)$ with $\rho = 1$, $\alpha = -2$. Correspondingly the substitution $u = 1 - 2x$ conjugates f_2 to the monomial $u \mapsto u^2$, and the linearised fixed point has multiplier 2 (repelling), as theorem 3.12(i) requires. For $r = 4$, inverting (3.31): $t = (\arcsin \sqrt{x})^2$ gives

$$\varphi_4(t) = \psi_4^{-1}(t) = \sin^2 \sqrt{t} = \frac{1 - \cos(2\sqrt{t})}{2},$$

a Möbius image of $\cos(\alpha t^\rho + \beta)$ with $\rho = \frac{1}{2}$, $\alpha = 2$, $\beta = 0$ — the fractional exponent is admissible precisely because $\cos(2\sqrt{t})$ is an entire function of t . Correspondingly $u = 1 - 2x$ conjugates f_4 to $T_2(u) = 2u^2 - 1$, the second Chebyshev polynomial, and the multiplier at the linearised fixed point is 4 (repelling).

For $r \in (0, 1)$ the fixed point is attracting, no member of the classification is available, and theorem 3.14 gives hypertranscendence of ψ_r — which underpins the claims of this thesis for $A(p) = 2\Psi_{2p}(\frac{1}{2})$.

3.B Closed-form search: exploratory formulae

This appendix records the candidate formulae produced by the searches of [section 3.7](#). They are approximations only, quarantined here because they are not needed for the probabilistic role of $A(p)$ in the main text; they are retained because they show how little a visually accurate fit says about the singular quantity $1/A(p)$. These expressions have no inferential status: they are records of a curve-fitting exercise, not proposed identities.

3.B.1 Rational candidates

Polynomial and rational searches can be constrained to the known endpoints $A(0) = \frac{1}{2}$ and $A(\frac{1}{2}) = 0$. One of the more economical rational candidates was

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}, \quad (3.32)$$

the candidate $\hat{A}_1$ of [section 3.7](#). It passes through both endpoints. At the critical endpoint, however,

$$\hat{A}'_1\left(\frac{1}{2}\right) = -\frac{2}{0.55} \approx -3.636,$$

whereas [theorem 3.4](#) requires the left derivative -4 . Matching the values but missing this slope is enough to make the relative error in A , and hence the error in $1/A$, unacceptable near criticality.

3.B.2 Tree-search candidates

A less restricted symbolic tree search over progressively nested elementary functions produced many longer formulae of similar visual quality. Two representative outputs were

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \ln \left(\cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \\ & \left. \left. + p \left[\sin \left(\cos(|p|(0.621865696665606 - p) \cos(e^{e^p})) \right) + |p| \right] \right) \right) + 0.5982282661889831 \end{aligned} \quad (3.33)$$

and the comparatively concise

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771. \quad (3.34)$$

Redundancies such as the unsimplified p/p occur in raw outputs of the search; they are introduced by the programme much like neutral mutations in biological evolution, having no effect on fitness, and [\(3.33\)](#) shows the same expression with them removed.

3.B.3 Splicing the critical branch

One can force the correct critical branch by splicing the asymptotic onto a fitted curve:

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases} \quad (3.35)$$

This is a practical surrogate, not a closed form for A . If a splice is needed, the certified product of [\(3.9\)](#) and [\(3.10\)](#) is preferable on the first branch and the proved asymptotic [theorem 3.4](#) on the second; that is the construction adopted in [section 3.11](#).

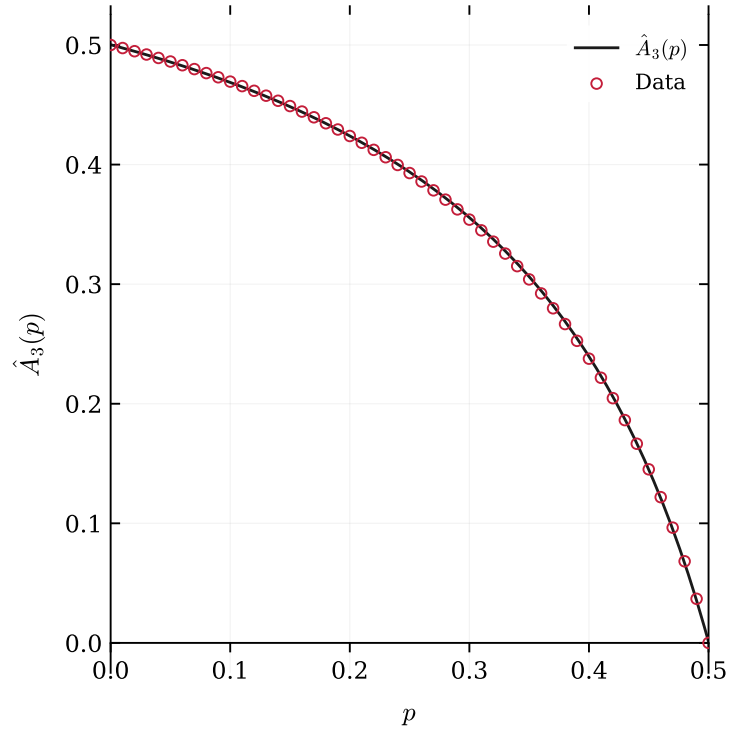


Figure 3.10: The fitted curve $\hat{A}_3$ of (3.34) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) \approx 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient approximately -3.63 instead of -4 . Errors of this size are invisible here but are fatal for $1/A$ near criticality.

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